Why Do Deep Neural Networks Generalize Well?

From the topological and geometrical perspectives

Implicit Biases of Deep Neural Networks

Classic statistical learning theory establishes a trade-off between bias and variance, which suggests that an overparameterized model would generalize poorly to unseen data. However, modern deep neural networks generalize well (low generalization error) despite being overparameterized (near-zero training error) [1, 2], thus contradicting the traditional statistical wisdom. This phenomenon is referred to as *benign overfitting*, which can also occur in non-deep learning models, such as trees, boosting and linear regression [3, 4]. In particular, the generalization error exhibits a *double descent* pattern in relation to the number of parameters, instead of the classic U-shaped convex pattern, where the additional second descent in generalization error usually occurs at the transition to the overparameterized regime [3, 5, 6]. This necessitates a reexamination of the relation between training and generalization, which has led to many interesting and engaging discussions in the last few years [5, 7, 8, 9, 10]. However, these works attempting to explain benign overfitting all have shortcomings, common of which is that they mostly validate their hypothesis empirically on small-sample regimes and mostly focus only on one aspect, thus lacking a unified view on the phenomenon.

The existing attempts to explain this phenomenon can be classified into two groups. The first group proposes new model complexity measures which are much smaller than the number of parameters within the model, and thus argues that overparameterized models appear more complex than they really are. A representative work in this group is [5], which argues that the double descent pattern is the result of a 1-dimensional representation of a 2-dimensional phenomenon.

The second group hypothesize that deep neural networks have implicit bias towards functions that generalize well [11, 12]. In particular, the functions that generalize well are presumed to have low complexity. What are the resources of this hypothesized bias? There are two prevailing theories, one arguing that this bias originates from the loss landscape or the architecture (i.e., the parameter-function mapping) of the model [7] and the other arguing that it's the properties of the gradient-based stochastic optimization algorithm, like stochastic gradient descent (SGD) [13, 14]. Existing works are limited in disentangling the source of this bias because they only provide fragmented, sometimes contradictory, insights.

The key idea for this proposal is to use tools from geometry and topology to help explain the reasons behind the generalization property of deep neural networks. The goal of this line of pursuit is to provide a theoretical examination and a unified view of the phenomenon.

Topology and Geometry in Neural Networks

From my perspective, topology and geometry are valuable tools for gaining insights into the generalization of deep neural networks and are underutilized in the existing literature. Here, I will summarize the key insights gained using tools from topology and geometry so far, some shortcomings within the existing literature and highlight what more could be done.

Several works have been using persistent homology (from algebraic topology) to examine the topological structural changes of the data as it passes through the layers in well-trained deep neural networks (i.e., near-zero training error) [15, 16, 17]. They demonstrate *empirically* 1) why nonsmooth activation like ReLU performs better than the smooth one, and 2) why deep neural networks often perform better than shallow ones. The Key insight here is that the topological complexity of the data gets progressively simpler as it moves through the layers. A smooth function preserves the data's topological features and thus is less effective in reducing the topological complexity which explains the performance gain from nonsmooth activations. Another interesting observation is that deep neural networks spread the simplification of the topological features across multiple layers, while shallow ones reduce the topological complexity at the last layer instead. This is rather unsatisfactory, as both deep and shallow neural networks simplify the topological structure at the end, which suggests topological simplications do not sufficiently explain the performance gain of deep neural networks over shallow ones. Furthermore, these works focused only on examining the topological structures of the trained neural networks and did not attempt to explain the reasons behind the generalization property of the overparameterized neural networks.

Following the line of topological analysis, [18, 19, 20] examine the topological structure of the training trajectory and illustrate that generalization error can be equivalently bounded in terms of a topological feature called the *persistent homology dimension* (PHD) [21]. Furthermore, they demonstrate *empirically* that this PHD is predictive of the generalization error for deep networks. This is an interesting observation, which suggests that the underlying topology of the training dynamics impacts the generalization. However, it does not contribute much in terms of disentangling the source of the bias, gradient-based optimization algorithms vs loss landscape, as all the experiments are done using some variants of the SGD algorithms.

Switching to the geometry perspective, a key theory, which utilizes algebraic geometry to analyze singular statistical models, including deep neural networks, is the *Bayesian singular learning theory* [22, 23, 24]. Intuitively, deep neural networks are singular in the sense that the parameter-to-function map is not injective [25]. This noninjectivity induces *singularities* in the parameter space, where the classical statistical learning fails as the Fisher information matrix is no longer positive-definite. Bayesian singular learning theory proposes a new complexity measure, called the *real log canonical threshold* (RLCT), which gives a new way of counting the effective number of parameters in a deep neural network by measuring the complexity of a resolution of its singularities. It establishes a new trade-off between the accuracy (fit to the training data) and complexity in terms of RLCT, and demonstrates that singular models generalize well due to low RLCT.

The major shortcomings of this theory are twofold. First, the assumption of an analytical loss landscape is crucial, which fails to hold for nonsmooth activations such as ReLU. Second, the assumption of realizability, i.e., there exists some parameter which achieves the true and unknown distribution.

Research Plan

To provide a theoretical examination and a unified view of the generalization of deep neural networks, this research plan will be divided into three steps. The first step will provide a theoretical examination of the geometrical and topological properties of the gradient-based optimization algorithm, SGD in particular. From the geometrical aspect, I will examine how SGD will explore the singularities within the parameter space. Furthermore, analyze the topology of these singularities can be a fruitful direction as well.

Other Minor Topics

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